MAUS User Guide

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> **Note:** This is a placeholder ("dummy") guide scaffold. Replace the bracketed `[TODO]` sections with project-specific content before publishing.

Methyl Assignments Using Satisfiability — a clean-room reimplementation of the SAT-based methyl NMR assignment method of Nerli et al., *Nat. Commun.* 12:691 (2021).

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1. Overview

MAUS assigns methyl NMR resonances by **subgraph isomorphism** rather than by scoring. Given a protein structure and real **peak lists** (HMQC + NOESY), it returns — for each peak — the *set* of methyls consistent with every hard constraint. It never emits a single guess unless the data force one, and it provably never excludes a valid assignment.

Concept	In this tool
Structure graph **G**	one node per methyl carbon; geminal / short / long edges
Data graph **H** nodes	HMQC peaks: (^{1}H, ^{13}C) shift + residue type
Data graph **H** edges	NOESY cross peaks, endpoints matched to HMQC by frequency
Solver	`python-sat` (Glucose3), assumption-based enumeration
Output	per-peak option set (1 / 2–3 / >3 / unassigned)

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2. Installation

```
python3 -m pip install python-sat
git clone https://github.com/deepnmr/maus.git
cd maus
```

Requirements:

- Python ≥ 3.8
- `python-sat` (provides the Glucose3 SAT solver)

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3. Quick start

```
# 1. build peak lists from a BMRB shift file + the PDB
python make_peaklists.py examples/mbp/1ANF.pdb
        examples/mbp/bmr7114_3.str

# 2. run MAUS on the HMQC + NOESY peak lists (--truth scores against
        the key)
python maus.py examples/mbp/1ANF.pdb examples/mbp/hmqc.tsv
        examples/mbp/noesy.tsv \
    --truth examples/mbp/hmqc_true.tsv --tol-h 0.01 --tol-c 0.05 --out
        mbp_options.tsv
```

Expected summary:

```
methyls(G nodes)=192  HMQC peaks=192  NOESY cross peaks=1650
NOE match: firm=502 ambiguous(dropped)=1148 unmatched=0
unique(1 option)      = 51/192
ambiguous(2-3 options)= 81/192
ambiguous(>3 options) = 60/192
unassigned            = 0/192
truth in option set   = 192/192 = 100.0%
```

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4. Input formats

4.1 PDB structure

Standard `ATOM` records. Only these residue types are parsed by default: `ALA ILE LEU MET THR VAL`. Chain `A` (or blank) is used.

4.2 HMQC peak list (TSV) — data-graph nodes (input)

```
label <TAB> H_ppm <TAB> C_ppm <TAB> res_type
P1      0.828        24.510       L
```

- `label` — anonymous peak id (`P1, P2, …`); leaks nothing about the answer
- `H_ppm`, `C_ppm` — methyl (^{1}H, ^{13}C) chemical shifts
- `res_type` — one-letter code (`A I L M T V`), known from labeling

Ground truth lives in a **separate** file, never in the input:

```
label <TAB> H_ppm <TAB> C_ppm <TAB> res_type <TAB> True
P1      0.828        24.510       L              L7CD1
```

Pass it with `--truth` to score; omit it to run blind.

4.3 NOESY peak list (TSV) — data-graph edges

```
label <TAB> C1 <TAB> C2 <TAB> H2
X1      24.56    25.39   0.340
```

3D `(H)CCH` methyl-methyl cross peak. The **observed** methyl is `(H2, C2)`; the NOE **partner** contributes carbon `C1` only (a 3D peak has no partner proton). During the run the observed methyl is matched to an HMQC peak by `(H2,C2)` and the partner by carbon `C1` alone (within `--tol-h`/`--tol-c`). Single distance class (structure edge ≤ `long_cut`). Because ^{13}C alone is degenerate, most cross peaks match several candidate partners and drop as ambiguous — carbon tolerance is the main resolution lever.

4.3b HMBC-HMQC peak list (TSV, optional) — geminal links

```
label <TAB> C1 <TAB> C2 <TAB> H2
B1      24.56    25.39   0.340
```

Same layout as the NOESY list. One row per Leu/Val residue: observed methyl `(H2, C2)`, geminal partner carbon `C1`. Pass with `--hmbc`; MAUS forces the pair onto a geminal structure edge. The carbon-only partner is degenerate (on MBP ~1/50 links resolve), and even a firm link couples the pair without fixing which residue — so per-peak option *counts* are essentially unchanged. The never-exclude guarantee is preserved.

4.4 Generating the peak lists

`make_peaklists.py` builds all three files from a PDB and a BMRB NMR-STAR shift file (`bmrXXXX_3.str`): `hmqc.tsv` (input), `hmqc_true.tsv` (truth key), and `noesy.tsv`. HMQC (^{1}H,^{13}C) coordinates are the real BMRB methyl shifts; NOESY cross peaks are emitted for structurally close methyl pairs.

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5. Command-line options

Option	Default	Meaning
`--short-cut`	6.0	structure short-range edge cutoff (Å)
`--long-cut`	10.0	structure long-range edge cutoff (Å)
`--tol-h`	0.02	^{1}H NOESY/HMBC→HMQC match tolerance (ppm)
`--tol-c`	0.20	^{13}C NOESY/HMBC→HMQC match tolerance (ppm)
`--hmbc`	–	optional HMBC-HMQC geminal-link peak list
`--truth`	–	truth-key TSV for scoring
`--labeling`	`A;I;L;M;T;V`	residue types present
`--out`	–	write per-peak options TSV

`make_peaklists.py` options: `--noe-short` (6.0), `--noe-long` (8.0), `--keep-k` (12 nearest NOE partners per methyl).

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6. Interpreting output

The `--out` TSV columns:

Column	Meaning
`label`	input peak identifier (anonymous)
`res_type`	one-letter residue type
`n_options`	number of valid methyls
`options`	comma-separated methyl labels
`truth`	ground-truth label (blank if no `--truth`)
`truth_in_set`	1 if truth ∈ options, else 0 (blank if no `--truth`)

- **1 option** → uniquely assigned.
- **2–3 options** → residual ambiguity, usually geminal/local symmetry.
- **>3 options** → weak local NOE network.
- **0 options** → over-constrained inputs (check cutoffs).

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7. On the bundled benchmark

MAUS sees only the two peak lists. HMQC (^{1}H,^{13}C) coordinates are the real BMRB 7114 shifts, and each NOESY endpoint is resolved back to an HMQC peak **by frequency**, not by identity — so the assignment is not circular and shift degeneracy has real consequences.

- `truth in option set = 100%` is the MAUS **guarantee**: a valid assignment is provably never excluded (not a measurement — a property of the exact enumeration).
- The residual ambiguity is **real** and dominated by the 3D format: the NOE partner is matched on ^{13}C only, which is degenerate, so most cross peaks match several candidate partners and drop. Carbon tolerance is the main lever (unique 29 → 51 → 70 /192 at `--tol-c` 0.1 / 0.05 / 0.02). A 4D experiment resolving the partner proton would recover far more.

The NOESY cross peaks themselves are still *generated from the structure* (close methyl pairs), so this remains a controlled benchmark rather than a picked experimental NOESY. Swap in a real NOESY peak list to go fully experimental — the `maus.py` interface does not change.

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8. Troubleshooting

Symptom	Likely cause	Fix
`ModuleNotFoundError: pysat`	solver not installed	`pip install python-sat`
many `unassigned` peaks	structure cutoffs too tight	raise `--long-cut`

Symptom	Likely cause	Fix
most peaks ambiguous	too many NOEs dropped as ambiguous	tighten `--tol-h`/`--tol-c`
high `unmatched` NOE count	tolerance too tight vs shift precision	loosen `--tol-h`/`--tol-c`

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9. References

- Nerli, De Paula, McShan & Sgourakis. *Backbone-independent NMR resonance assignments of methyl probes in large proteins*. **Nat. Commun.** 12:691 (2021). doi:10.1038/s41467-021-20984-0

- See `COMPARISON.md` for the head-to-head against MAGIC.

[TODO] Add author, license, and contact sections before release.